SMITH STREET CONNECTION ROAD

Traffic Impact Assessment Report

Location:	-
Date Prepared:	13/05/2026
Prepared By:	Planner's Name
CC Number:	CC0000

Table of Contents

1. Executive Summary

2. Design & Traffic Inputs

Analysis Parameters

Selected Site Details

3. Traffic Analysis & Results

Summary of Computed Results

4. Detour Analysis

Detour Route 1

5. Conclusion

List of Tables

Table 1: Analysis Parameters

Table 2: Selected Site Details

Table 3: Summary of Computed Results

Table 4: GROUPED DIRECTIONAL SUMMARY - D1 - 🟢 DIRECTION 1, GAZETTAL, EASTBOUND

Table 5: GROUPED DIRECTIONAL SUMMARY - D2 - 🔴 DIRECTION 2, AGAINST GAZETTAL, WESTBOUND

Table 6: DIRECTIONAL QUEUE SUMMARY - D1 - 🟢 DIRECTION 1, GAZETTAL, EASTBOUND

Table 7: DIRECTIONAL QUEUE SUMMARY - D2 - 🔴 DIRECTION 2, AGAINST GAZETTAL, WESTBOUND

Table 8: DIRECTIONAL QUEUE SUMMARY - SWT - 🟠 SWT QUEUE SUMMARY

Table 9: DIRECTIONAL VCR SUMMARY - D1 - 🟢 DIRECTION 1, GAZETTAL, EASTBOUND

Table 10: DIRECTIONAL VCR SUMMARY - D2 - 🔴 DIRECTION 2, AGAINST GAZETTAL, WESTBOUND

Table 11: TABLE 8

Table 12: TABLE 9

Table 13: TABLE 10

Table 14: TABLE 21

Table 15: TABLE 22 - PRECISION DRIVE🧭 D1 -> PRI D1

Table 16: TABLE 23 - PRECISION DRIVE🧭 D1 -> PRI D1

Table 17: TABLE 24 - PRECISION DRIVE🧭 D1 -> PRI D1

Table 18: TABLE 25 - PRECISION DRIVE🧭 D1 -> PRI D1

Table 19: Detour Route 1

Table 20: 1. DETOUR ROUTE INFORMATION

Table 21: 2. ESTIMATED DETOUR DELAY CALCULATION

Table 22: ESTIMATED DELAY - DETOUR ROUTE

Table 23: 3. PEDESTRIAN DETOUR IMPACT

Table 24: PEDESTRIAN DETOUR IMPACT INCLUDE IN PRINT: YES - ESTIMATED DELAY - DETOUR ROUTE (PRECISION DRIVE >> SMITH STREET MOTORWAY)

1. Executive Summary

This Traffic Impact Assessment relates to SMITH STREET CONNECTION ROAD at -. The assessment has been prepared for 13/05/2026 using the current TWO-WAY operating configuration and focuses on queue performance, directional demand balance, and whether mitigation or diversion is warranted under the modeled condition.

Traffic demand context indicates that the modeled daily traffic volume is approximately 77,675 vehicles per day at opening year (2026), directional demand is split between D1 (37,096 vpd, 47.8% of total) and D2 (40,579 vpd, 52.2% of total), an annual growth rate of 2.50% has been applied (2024 → 2026).

Operational performance outcomes show that the worst modeled V/C ratio is 2.42 (LOS F); the peak queue demand is approximately 1,440 m (e.g. 1,440 m queue for 2-minute hold); directional peak queues are D1 4,343 m and D2 5,396 m.

On the basis of the current queue and capacity outputs, detour planning should be treated as an active mitigation requirement. Key supporting observations include: High Risk Period: AM • Network Max Queue ≈ 5,396 m; High Risk Period: AM • Worst VCR = 2.358 (LOS F); Pedestrian: Existing route 83.34s, detoured route 333.34s, crossing delay 120s.

Click here to refine the executive narrative, project-specific implications, and stakeholder-facing conclusions.

2. Design & Traffic Inputs

Analysis Parameters

Analysis parameters capture the selected site, growth, and directional demand assumptions used for the opening-year assessment. The assessment is anchored to Site ID 11545; SMITH STREET CONNECTION ROAD; count year 2024. The selected site reports HV content of N/A. Applied heavy-vehicle and rigid-truck inputs are D1 HV 7.6%, D2 HV 7.6%, D1 RT 0.15%, D2 RT 0.15%. A Google Maps reference is included in the Selected Site Details section for field verification.

> **Detailed table below sets out the supporting values used for the engineering interpretation.**

ROAD OPERATION MODE	TWO-WAY
BASE YEAR	2024
OPENING YEAR	2026
BASE YEAR AADT	73931
OPENING YEAR AADT	77675
D1 VADT OPENING YEAR	37096
D2 VADT OPENING YEAR	40579
GROWTH RATE PERCENT	2.5

These inputs should be checked first during review because any change to the selected site source data, opening year, growth rate, HV share, or RT share will flow directly through the hourly profile, queue, and V/C outputs.

Selected Site Details

Selected site details record the source counter, field description, mapping reference, and the heavy-vehicle / rigid-truck assumptions that feed the assessment.

> **Detailed table below confirms the selected counter, map reference, and applied vehicle-mix inputs used in the report.**

SITE ID	11545
SOURCE	TMR
ROAD NAME	SMITH STREET CONNECTION ROAD
DESCRIPTION	East of Precision Dr
COORDINATES	-27.963128, 153.363311
GOOGLE MAPS	View on Google Maps
COUNT YEAR	2024
APPLIED GROWTH RATE	N/A
SELECTED SITE HV%	N/A
DATA QUALITY	Measured (Hourly)
D1 DIRECTION	Eastbound
D2 DIRECTION	Westbound
D1 VADT	35,308
D2 VADT	38,623
TOTAL VADT	73,931
APPLIED D1 HV%	7.6

APPLIED D2 HV%	7.6
APPLIED D1 RT%	0.15
APPLIED D2 RT%	0.15

Use this section to verify that the adopted counter, mapped location, and HV / RT settings match the corridor actually being assessed before relying on the downstream queue or V/C outputs.

3. Traffic Analysis & Results

Summary of Computed Results

Summary of Computed Results sets out the 4-row result dataset covering Queue Peak M, Worst Vcr, Los. The highest recorded value is 5,396.00 for Queue Peak M (Value).

> **Detailed table below sets out the computed metrics with context on when each controlling condition occurs.**

METRIC	VALUE	*Context / When*
QUEUE PEAK M	5396	*Maximum queue length based on a standard 2-minute hold and release cycle*
WORST VCR	2.42	*Significantly over capacity (LOS F) — urgent mitigation required*
LOS	F	*Level of Service — A=free-flow, F=breakdown*
DETOUR RECOMMENDED	True	—

These controlling outputs typically emerge when directional demand, lane closure constraints, and work-zone control delays align during the peak operating period. A high queue or V/C outcome usually signals that short disturbances, heavy-vehicle platoons, or an extended stop-go cycle can rapidly push the corridor into unstable operation.

GROUPED DIRECTIONAL SUMMARY - D1 - 🟢 DIRECTION 1, GAZETTAL, EASTBOUND

The directional traffic summary covers 2024, 2026, Hrly Avg / lane. The highest recorded value is 37,096 for 2026 in the HV column. Values range from 2 to 37,096 across all periods.

> **Detailed table below sets out the supporting values used for the engineering interpretation.**

YEAR	AM LV	AM HV	AM RT	OP LV	OP HV	OP RT	PM LV	PM HV	PM RT	EV LV	EV HV	EV RT	TOTAL
2024	1,889	156	4	2,147	178	4	2,654	219	5	2,335	193	4	35,308
2026	1,985	164	4	2,257	186	4	2,789	230	5	2,453	203	4	37,096
Hrly Avg / lane	993	82	2	1,129	93	2	1,395	115	3	1,227	102	2	-

The conditions reflected in this table usually become important when multiple moderate effects combine within the same operating window. Review the controlling rows and periods together rather than in isolation so the mitigation response remains aligned with the actual operational trigger.

GROUPED DIRECTIONAL SUMMARY - D2 - 🔴 DIRECTION 2, AGAINST GAZETTAL, WESTBOUND

The directional traffic summary covers 2024, 2026, Hrly Avg / lane. The highest recorded value is 40,579 for 2026 in the HV column. Values range from 2 to 40,579 across all periods.

> **Detailed table below sets out the supporting values used for the engineering interpretation.**

YEAR	AM LV	AM HV	AM RT	OP LV	OP HV	OP RT	PM LV	PM HV	PM RT	EV LV	EV HV	EV RT	TOTAL
2024	3,327	275	6	2,696	223	5	2,207	182	4	2,170	179	4	38,623
2026	3,496	289	6	2,833	234	5	2,318	192	4	2,280	188	4	40,579
Hrly Avg / lane	1,748	145	3	1,417	117	3	1,160	96	2	1,141	94	2	-

The conditions reflected in this table usually become important when multiple moderate effects combine within the same operating window. Review the controlling rows and periods together rather than in isolation so the mitigation response remains aligned with the actual operational trigger.

DIRECTIONAL QUEUE SUMMARY - D1 - 🟢 DIRECTION 1, GAZETTAL, EASTBOUND

Directional Queue Summary - D1 - 🟢 Direction 1, Gazettal, Eastbound sets out the 5-row result dataset covering ★ 1 Lane Closed (1 open) — Per 5 min, Per 2 min, Per 5 min. The highest recorded value is 8,686.00 for Per 15 min (PM).

> **The table below provides queue length results by time period and direction.**

QUEUE DURATION	AM	OP	PM	EV	MAX QUEUE
★ 1 Lane Closed (1 open) — Per 5 min	2,104	2,380	2,896	2,584	2,896
Per 2 min	842	952	1,158	1,034	1,158
Per 5 min	2,104	2,380	2,896	2,584	2,896
Per 10 min	4,206	4,758	5,790	5,166	5,790
Per 15 min	6,310	7,138	8,686	7,750	8,686

Elevated queue conditions commonly occur when temporary traffic control phases, reduced lane capacity, or strong directional peaks limit discharge opportunities. This is the table to review when assessing whether queue storage may extend into upstream accesses, intersections, or sensitive frontages.

DIRECTIONAL QUEUE SUMMARY - D2 - 🔴 DIRECTION 2, AGAINST GAZETTAL, WESTBOUND

Directional Queue Summary - D2 - 🔴 Direction 2, Against Gazettal, Westbound sets out the 5-row result dataset covering ★ 1 Lane Closed (1 open) — Per 5 min, Per 2 min, Per 5 min. The highest recorded value is 10,792.00 for Per 15 min (AM).

> **The table below provides queue length results by time period and direction.**

QUEUE DURATION	AM	OP	PM	EV	MAX QUEUE
★ 1 Lane Closed (1 open) — Per 5 min	3,598	2,932	2,416	2,398	3,598
Per 2 min	1,440	1,174	966	960	1,440
Per 5 min	3,598	2,932	2,416	2,398	3,598
Per 10 min	7,194	5,862	4,830	4,794	7,194
Per 15 min	10,792	8,794	7,246	7,192	10,792

Elevated queue conditions commonly occur when temporary traffic control phases, reduced lane capacity, or strong directional peaks limit discharge opportunities. This is the table to review when assessing whether queue storage may extend into upstream accesses, intersections, or sensitive frontages.

DIRECTIONAL QUEUE SUMMARY - SWT - 🟠 SWT QUEUE SUMMARY

Directional Queue Summary - SWT - 🟠 SWT Queue Summary sets out the 3-row result dataset covering Direction 2, Against Gazettal, Westbound (Red 120s), Direction 1, Gazettal, Eastbound (Red 120s), ⚠️ SWT input anomaly (demand capped to sat-flow): Direction 1, Gazettal, Eastbound 13/24 hour(s), Direction 2, Against Gazettal, Westbound 14/24 hour(s). The highest recorded value is 278.00 for Direction 2, Against Gazettal, Westbound (Red 120s) (AM).

> **The table below provides queue length results by time period and direction.**

DIRECTION	AM	OP	PM	EV	MAX SWT QUEUE
Direction 2, Against Gazettal, Westbound (Red 120s)	278	278	278	278	278
Direction 1, Gazettal, Eastbound (Red 120s)	278	278	278	278	278
⚠️ SWT input anomaly (demand capped to sat-flow): Direction 1, Gazettal, Eastbound 13/24 hour(s), Direction 2, Against Gazettal, Westbound 14/24 hour(s)					

Elevated queue conditions commonly occur when temporary traffic control phases, reduced lane capacity, or strong directional peaks limit discharge opportunities. This is the table to review when assessing whether queue storage may extend into upstream accesses, intersections, or sensitive frontages.

DIRECTIONAL VCR SUMMARY - D1 - 🟢 DIRECTION 1, GAZETTAL, EASTBOUND

*Directional VCR Summary - D1 - 🟢 Direction 1, Gazettal, Eastbound sets out the 4-row result dataset covering Design Capacity * = 1,568 vph/lane (3,136 vph directional), Base VCR, ★ 1 Lane Closed (1 open). The highest recorded value is 1.88 for ★ 1 Lane Closed (1 open) (PM).*

> **The table below provides volume-to-capacity ratio and Level of Service results by time period.**

METRIC	AM	OP	PM	EV
Design Capacity * = 1,568 vph/lane (3,136 vph directional)	-	-	-	-
Base VCR	0.67 LOS B ≤0.70	0.69 LOS B ≤0.70	0.94 LOS E ≤1.00	0.32 LOS A ≤0.60
★ 1 Lane Closed (1 open)	1.34 LOS F >1.00	1.38 LOS F >1.00	1.88 LOS F >1.00	0.64 LOS B ≤0.70
Single Lane Reversible Flow	N/A	N/A	N/A	N/A

High V/C conditions occur when effective work-zone capacity is constrained by lane closure, heavy vehicles, turning friction, or peak directional loading. Values near or above practical capacity indicate the network has limited resilience, so relatively minor disturbances can trigger disproportionate delay and recovery time.

DIRECTIONAL VCR SUMMARY - D2 - 🔴 DIRECTION 2, AGAINST GAZETTAL, WESTBOUND

*Directional VCR Summary - D2 - 🔴 Direction 2, Against Gazettal, Westbound sets out the 4-row result dataset covering Design Capacity * = 1,568 vph/lane (3,136 vph directional), Base VCR, ★ 1 Lane Closed (1 open). The highest recorded value is 2.36 for ★ 1 Lane Closed (1 open) (AM).*

> **The table below provides volume-to-capacity ratio and Level of Service results by time period.**

METRIC	AM	OP	PM	EV
Design Capacity * = 1,568 vph/lane (3,136 vph directional)	-	-	-	-
Base VCR	1.18 LOS F >1.00	0.82 LOS D ≤0.90	0.80 LOS C ≤0.80	0.29 LOS A ≤0.60
★ 1 Lane Closed (1 open)	2.36 LOS F >1.00	1.65 LOS F >1.00	1.59 LOS F >1.00	0.58 LOS A ≤0.60
Single Lane Reversible Flow	N/A	N/A	N/A	N/A

High V/C conditions occur when effective work-zone capacity is constrained by lane closure, heavy vehicles, turning friction, or peak directional loading. Values near or above practical capacity indicate the network has limited resilience, so relatively minor disturbances can trigger disproportionate delay and recovery time.

4. Detour Analysis

This section summarises the detour route analysis. For each alternative route, the road capacity, estimated motorist delay, and pedestrian impact are assessed.

Detour Route 1

1. DETOUR ROUTE INFORMATION

PARAMETER	DETAILS
Detour Route Name	**Precision Drive >> Smith Street Motorway**
Selected Path	**Precision Drive >> Smith Street Motorway**
Route Length (km)	5km
Road Classification	Urban

Edit to insert the selected detour route and path details.

2. ESTIMATED DETOUR DELAY CALCULATION

ESTIMATED DELAY - DETOUR ROUTE

Estimated Delay - Detour route sets out the 6-row result dataset covering Detour Route Length (km), Average Travel Speed on Detour (km/h), Travel Time (seconds). The highest recorded value is 70.00 for Average Travel Speed on Detour (km/h) (Value).

> **The table below provides supporting values for the detour or pedestrian impact assessment.**

PARAMETER	VALUE
Detour Route Length (km)	0.4
Average Travel Speed on Detour (km/h)	70
Travel Time (seconds)	21
Total Time incl. Intersections (seconds)	61
Estimated Additional Travel Time (min)	2
Delay Classification	Moderate (1-5 min)

Detour pressure becomes material when diverted traffic is reassigned to roads with limited spare capacity during the same peak period as the primary corridor impact. These conditions are most important where side-road intersections, local access, or school and freight activity are already operating close to their normal thresholds.

3. PEDESTRIAN DETOUR IMPACT

PEDESTRIAN DETOUR IMPACT INCLUDE IN PRINT: YES - ESTIMATED DELAY - DETOUR ROUTE (PRECISION DRIVE >> SMITH STREET MOTORWAY)

Pedestrian Detour Impact Include in Print: Yes - Estimated Delay - Detour route (Precision Drive >> Smith Street Motorway) sets out the 1-row result dataset covering 83.34. The highest recorded value is 370.00 for 83.34 (Added Delay (s)).

> **The table below provides supporting values for the detour or pedestrian impact assessment.**

EXISTING ROUTE TIME (S)	DETOURED ROUTE TIME (S)	CROSSING DELAY (S)	ADDED DELAY (S)	ADDED DELAY (MIN)
83.34	333.34	120	370	6.17

Detour pressure becomes material when diverted traffic is reassigned to roads with limited spare capacity during the same peak period as the primary corridor impact. These conditions are most important where side-road intersections, local access, or school and freight activity are already operating close to their normal thresholds.

5. Conclusion

The assessment for SMITH STREET CONNECTION ROAD at - indicates that the modeled road network should be interpreted as a constrained work-stage environment rather than a normal operating condition with approximately 77,675 vehicles per day and a growth allowance of 2.50% per annum. The reported outputs are therefore most useful as a screening tool for operational risk, queue storage exposure, and the timing of mitigation triggers.

Operationally, the model points to a worst V/C ratio of 2.42 corresponding to LOS F and a peak queue on the order of 1,440 m (2-minute hold cycle). These values suggest that traffic performance is being controlled by a limited number of peak-period constraints, so site management should focus on those controlling periods rather than average daily conditions.

From an engineering perspective, mitigation should prioritise maintaining discharge opportunities, reducing unnecessary blockage time, and protecting upstream intersections and accesses from queue spillback. Where detour planning is triggered, the diversion strategy should be treated as an operational management measure with active monitoring rather than a one-off desktop assumption.

- Detour planning should be retained as an active mitigation measure because the modeled condition indicates insufficient resilience under the controlling scenario.
- The controlling risk is concentrated in peak operating periods rather than uniformly across the day.
- Queue and V/C results should be used to set monitoring triggers, traffic controller response actions, and escalation points for temporary traffic management.
- Any material change in closure duration, lane availability, or directional demand should prompt the assessment to be refreshed before implementation.
- The detailed table set has been interpreted together with 28 supporting result table(s) so that mitigation can be tied to the specific controlling scenario.

PREPARED BY

Planner's Name

Planner's Name

REVIEWED BY

Reviewer Name

Reviewer's Name

ENGINEERING DISCLAIMER: This Traffic Impact Assessment (TIA) report has been prepared based on site-specific observations and industry-standard modeling assumptions. While every effort has been made to ensure accuracy, real-world traffic conditions are subject to variables including driver behavior, weather, and external network incidents. This report is intended for use by planning authorities and should be reviewed in the context of the complete project documentation.